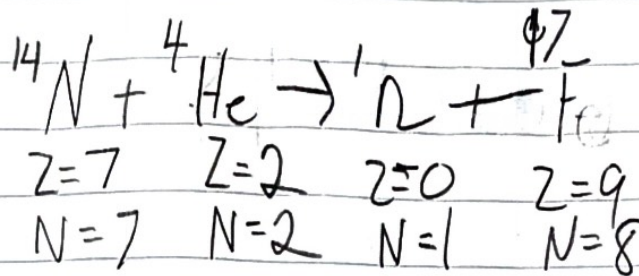


Homework #4

1.)



a.) minimum α -energy for ${}^{14}\text{N}$ target.

$$T_p \geq -Q \frac{m_p + m_T}{m_T}$$

$$\Delta({}^{14}\text{N}) = 2863.41683 \text{ keV}$$

$$\Delta({}^4\text{He}) = 2424.91587 \text{ keV}$$

$$\Delta({}^1\text{n}) = 8071.31806 \text{ keV}$$

$$\Delta({}^{17}\text{F}) = 1951.701 \text{ keV}$$

$$Q = (\Delta({}^{14}\text{N}) + \Delta({}^4\text{He}) - (\Delta({}^1\text{n}) + \Delta({}^{17}\text{F})))$$

$$Q = (2863.41683 + 2424.91587) - (8071.31806 + 1951.701)$$

$$Q = -4.7347 \text{ MeV}$$

$$m_\alpha = 4.00260325413$$

$$m_{{}^{14}\text{N}} = 14.00307400425$$

$$T_{\alpha} \geq \left(4.7347 \left(\frac{4.00260325413 + 14.00307400425}{14.00307400425} \right) \right)$$

$$T_{\alpha} \geq 6.0881 \text{ MeV}$$

b.)

$$T_{^{14}\text{Ne}} \geq \left(4.7347 \times \left(\frac{4.00260325413 + 14.00307400425}{4.00260325413} \right) \right)$$

$$T_{^{14}\text{Ne}} \geq 21.299 \text{ MeV}$$

2.) • Proton lab energy 6 MeV

• Proton capture cross-section for $p + {}^{14}\text{N}$

$$\sigma_{\text{total}} = \pi R^2 (1 - v_c/c)$$

$$R = 1.2 (14^{1/3} + 1^{1/3})$$

$$R = 4.092 \text{ fm}$$

$$V_c = \frac{Z_1 Z_2 e^2}{R} = \frac{(1)(7)(1.44 \text{ MeV-fm})}{(4.092 \text{ fm})}$$

$$V_c = 2.463 \text{ MeV}$$

$$\sigma_{\text{tot}} = \pi (4.092)^2 \left(1 - \frac{2.463 \text{ MeV}}{6 \text{ MeV}} \right) \left(\frac{60+30}{2} \right)$$

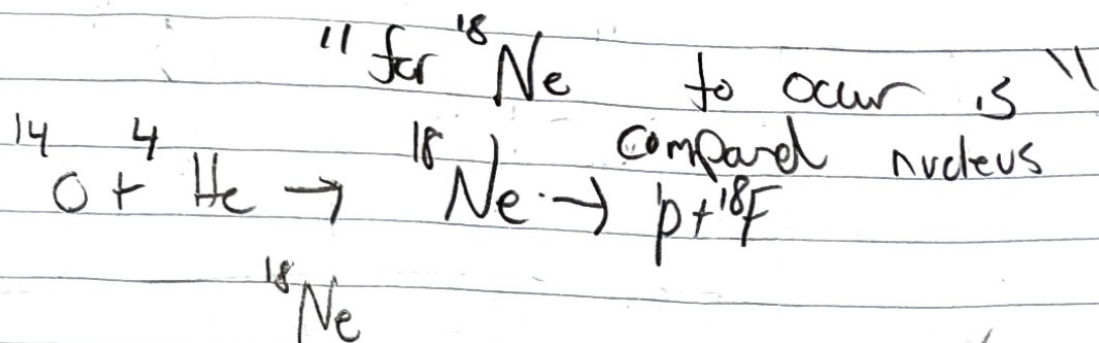
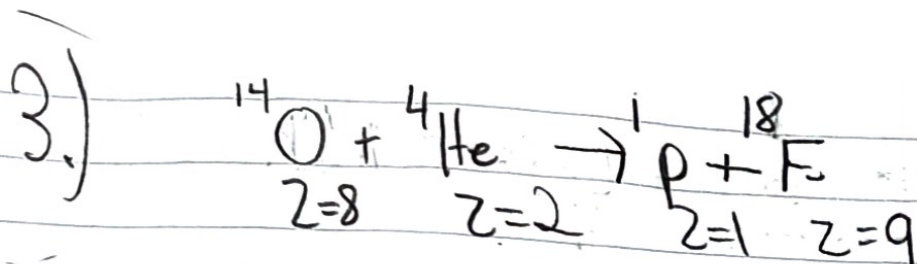
$$= 311.01 \text{ fm}^2 \left(\frac{10 \text{ mb}}{1 \text{ fm}^2} \right)$$

There is 1 order mag. diff from paper to result using semiclassical assumption.

$\sigma_{\text{tot}} \approx 311.01 \text{ mb}$ & $\sigma_{\text{paper}} = 45 \text{ mb}$

$\sigma_{\text{paper}} = 20-35 \text{ mb}$ (1.63 MeV curve)

$\sigma_{\text{paper}} = 50-60 \text{ mb}$ (2.31 MeV curve)



$^{18}_{10}\text{Ne}$
 $Z=10$ e-e nuclei
 $N=8$

so spin and parity types follow

$$\left| \left(l - \frac{1}{2} \right) \right| \leq J \leq \left| \left(l + \frac{1}{2} \right) \right|$$

$$\pi(l) = \pi_1 \pi_2 = (-1)^l$$

$l = 0, 1, 2, \dots$ $\uparrow$ $\uparrow$
 proj target

$$\pi(l=0) = +1$$

$$\pi(l=1) = -1$$

$$\pi(l=2) = +1$$

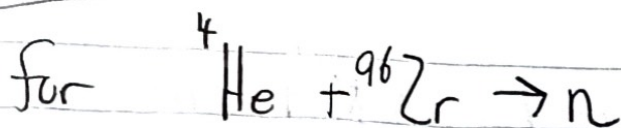
$$\pi(l=3) = -1$$

^{18}Ne is compound nucleus in resonant reaction then also e-e nuclei, then $\pi = +1, +1, +1, +1$

$$\pi(J) = (-1)^L$$

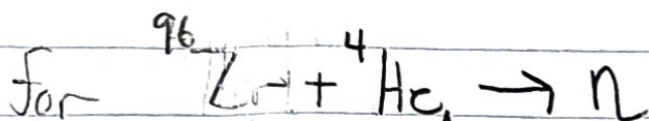
$^{19}\text{F} + n$ cross-section is $\sim 6h$

4.)



$$\begin{aligned} \text{Maximum } E_n &= 45,652.54 \text{ MeV} \\ \text{Minimum } E_n &= 45,520.628 \text{ MeV} \end{aligned}$$

$$E_n = 2.5 \text{ MeV}$$



$$\begin{aligned} \text{Maximum } E_n &= 45922.966 \text{ MeV} \\ \text{Minimum } E_n &= 45254.787 \text{ MeV} \end{aligned}$$

5) $^{197}\text{Au} + n$ cross-section is ~ 66

at $E_n = 500 \text{ keV}$

i) Estimate neutron-capture cross-section at $E_n = 100 \text{ keV}$

$$\sigma_{\text{total}} \propto \pi \lambda^2 \sqrt{E} \propto \pi \left(\frac{h^2}{2m_n E} \right) \sqrt{E} \propto \frac{1}{\sqrt{E}}$$

"Low energy total neutron interaction"

$$\sigma (66\text{-barn}) \propto \frac{1}{\sqrt{500 \text{ keV}}} \quad \sigma_{\text{total}} \propto \frac{1}{\sqrt{E}}$$

$$\sigma (X) \propto \frac{1}{\sqrt{100 \text{ keV}}}$$

$$\sigma(500 \text{ keV}) \propto \frac{1}{\sqrt{500 \text{ keV}}}$$

$$\sigma(100 \text{ keV}) \propto \frac{1}{\sqrt{100 \text{ keV}}}$$

$$\frac{\sigma(500 \text{ keV})}{\sigma(100 \text{ keV})} \propto \frac{\sqrt{100 \text{ keV}}}{\sqrt{500 \text{ keV}}}$$

$$\sigma_{nc} \approx \sigma \left(\frac{\sqrt{500 \text{ keV}}}{\sqrt{100 \text{ keV}}} \right) \text{ barns}$$

$$\sigma_{nc} \approx 13.42 \text{ barns}$$

Comparable to papers $\sim 11 \text{ barns}$

6.)

resonant (α, p) cross section.

$$\sigma = \frac{\pi}{k^2} g \frac{\Gamma_\alpha \Gamma_p}{(E - E_R)^2 + \frac{\Gamma_{tot}^2}{4}} = \frac{\pi g \Gamma_\alpha \Gamma_p}{k^2 (E - E_R)^2 + \frac{\Gamma_{tot}^2}{4}}$$

$$\sigma(E_R) = \frac{\pi g \Gamma_\alpha \Gamma_p}{k^2 \frac{\Gamma_{tot}^2}{4}}$$

$\Gamma_\alpha \gg \Gamma_p$ apply this condition

$$\Gamma_{tot} = \Gamma_\alpha + \Gamma_p = \Gamma_\alpha$$

$$\sigma(E_R) = \frac{\pi g \Gamma_\alpha \Gamma_p}{k^2 \frac{\Gamma_\alpha^2}{4}} \quad 4$$

$$\sigma(E_R) = \frac{4\pi g}{k^2} \frac{\Gamma_p}{\Gamma_\alpha}$$

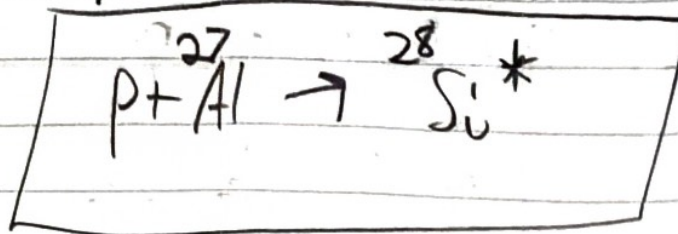
Γ_α width because $\Gamma_\alpha \gg \Gamma_p$
what greatly influence the
cross-section

7.)? What is the compound nucleus resulting from protons bombarding an aluminum target?

- Provide at least 5 different ways for compound nucleus to decay.

• Assume target is $^{27}_{13}\text{Al}$, beam is the proton p ($Z=1$)

Compound nucleus:



5-different ways for Decay:

- 1.) $^{28}_{14}\text{Si}^* \rightarrow ^{28}_{14}\text{Si} + \gamma$ ("gamma emission", radiative capture)
- 2.) $^{28}_{14}\text{Si}^* \rightarrow ^{27}_{13}\text{Al} + p$ ("proton emission", elastic/inelastic scattering)
- 3.) $^{28}_{14}\text{Si}^* \rightarrow ^{27}_{14}\text{Si} + n$ ("neutron emission", knockout)
- 4.) $^{28}_{14}\text{Si}^* \rightarrow ^{26}_{13}\text{Al} + \alpha$ ("alpha emission", knockout)
- 5.) $^{28}_{14}\text{Si}^* \rightarrow ^{26}_{12}\text{Mg} + d$ ("deuteron emission", transfer)

8.) (d,p) reaction ^{52}Cr leads to $\frac{3}{2}^-$ ground state of ^{53}Cr

- for direct reaction deuteron embeds/sheds neutron to ^{52}Cr nucleus then proton is emitted this neutron if it's in orbital with $l=1$ then ^{53}Cr can have gs with $\frac{3}{2}^-$ for J^π .
- for compound reaction it's inconclusive

when deuteron is absorbed and compound nucleus ^{54}Cr will only be able to describe exit channels based on statistical assumption meaning that the difference of not being able to be exact $J^\pi = \frac{3}{2}^-$ for that formalism.

("Bremsstrahlung")

Photo-nuclear reaction:

It's a reaction where high energy photon (X-ray) interacts with a nucleus and causes it to emit fragments. Main mechanism is Giant dipole resonance.

Giant dipole resonance occurs mechanism happens after photon hits a nucleus by causing the proton/neutron to oscillate in opposite directions. The ability to see that resonance is why confirms the mechanism is oscillatory.

Photons can be produced by for example one way is Bremsstrahlung, which happens when accelerated electrons hit a high-Z target the electric field of target deflects electron and from that deflection photons are emitted.

Paper example :

- Result of Photo nuclear cross sections
on ^{197}Au 1 cm plate of gold standard

basically measured cross sections
for reactions: $^{197}\text{Au}(\gamma, n)$ ^{196}Au and
 $^{197}\text{Au}(\gamma, 3n)$ ^{194}Au using

monenergetic gammas from range 13-31 MeV.
one of the results was that
 $^{197}\text{Au}(\gamma, n)$ ^{196}Au σ_{th} cross sections
measurements at approximately
beam energies 13-31 MeV.